

AGENTIC AI Engineer

PHASE-I

Core Mental Models.

- * LLM → probabilistic reasoning engine
- * Agents → control loops, not magic
- * AI Systems → distributed system with uncertainty
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- State Machine & Interoflows
- Async System
- API-first Architecture

Do not Touch LangChain or LangGraph.

PHASE-II

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Deep LLM + Retrieval Mastery.

In depth

- Tokenization & Context Optimization
- Embedding Spaces (How similarity Actually looks?)
- Advanced Rag:
 - Query Rewriting
 - Multi-hop Retrieval
 - Hybrid Search

Think about building $\rightarrow$ fast
 $\rightarrow$ accurate
 $\rightarrow$ Cheap

Retrieval Systems

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PHASE-III

Agent Architecture

Understand Core patterns:

$\rightarrow$ ReACT (Reason + Act loop)

$\rightarrow$ Plan $\rightarrow$ Execute $\rightarrow$ Reflect loop

$\rightarrow$ Tool Augmented learning

Frameworks

$\rightarrow$ LangGraph $\rightarrow$ controllable flows

$\rightarrow$ CrewAI $\rightarrow$ Role based Agent

$\rightarrow$ AutoGPT $\rightarrow$ Autonomous loops

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Multi-Agent Systems

Learn About

- Role specialization
- Communication protocols
- Task decomposition strategies.

Architectures:-

- Hierarchical Agents.
- Collaborative Agents.
- Competitive Agents

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⇒ Build Systems where Agents

→ debate

→ verify each other

→ Improve Output Quality.

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PHASE-V

Memory Systems.

Agents without memory = Toy project.

Types :-

↳ short-term (context window)

↳ long term (Vector DB)

↳ Episodic memory (past Actions)

↳ Semantic memory (Knowledge)

Go ~~deeper~~ Deeper :-

→ Memory Retrieval strategies.

→ context compression.

→ Forgetting mechanism.

PHASE-6

Tooling & Real World Action Layer.

→ Function Calling

→ API Orchestration

→ External tools

→ dB.

→ Browsers.

→ code execution

PHASE-VII

Planning, Reasoning & Self-Improvement

- Tree of Thoughts
 - Reflection loops
 - self-healing Agents
 - Goal decomposition.
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Build ~~new~~ Agents that:-

- detect mistakes
- retry intelligently
- improve over time.

PHASE-VIII

Evaluation & Reliability

- Eval pipelines
- Test datasets
- Failure tracking.

Learn.

- prompt regression testing
 - Agent benchmarking.
 - Cost vs accuracy tradeoffs.
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